

ONLINE RETAIL

Customer Analytics & Segmentation

An RFM & K-Means Clustering Analysis

Capstone Project — Business Report

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Source: UCI Machine Learning Repository — Online Retail Data Set

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1. Executive Summary

This report presents a customer analytics and segmentation study built on the UCI “Online Retail” dataset, a transaction-level record of a UK-based online gift retailer spanning December 2010 to December 2011. The purpose of the engagement was twofold: first, to translate raw transaction data into clear, decision-ready business insights about sales patterns, product performance, and customer geography; and second, to segment the retailer's customer base into behaviorally distinct groups so that marketing, retention, and account-management resources can be allocated more effectively.

The analysis followed a standard data science pipeline — data cleaning, exploratory data analysis (EDA), RFM (Recency, Frequency, Monetary) feature engineering, and unsupervised clustering via K-Means — to arrive at a three-segment customer model validated by a Silhouette Score of 0.594, indicating well-separated, cohesive clusters.

Business Value

- Identifies where revenue concentration exists (products, customers, and countries) so commercial teams can protect and grow the highest-value relationships.
- Quantifies seasonal and behavioral demand patterns to support inventory planning, staffing, and campaign timing.
- Produces an actionable three-tier customer segmentation — Champions, Active, and At Risk — that can be operationalized directly in CRM and email marketing platforms.
- Provides a data-driven foundation for retention strategy, loyalty investment, and win-back campaigns, replacing one-size-fits-all marketing with targeted, segment-specific action.

In short, the analysis shows that a small number of customers and products generate a disproportionate share of revenue, that sales activity is highly seasonal and time-of-day dependent, and that the customer base splits cleanly into a high-value “Champions” group, a stable “Active” middle tier, and a large “At Risk” group that requires proactive re-engagement.

2. Data Insights (Exploratory Data Analysis)

The raw dataset contained 541,909 transaction line items across 8 fields. Data cleaning removed rows with missing Customer IDs (135,080 rows, required for individual-level segmentation), 5,268 duplicate records, cancelled orders, and rows with non-positive quantity or price. The resulting analytical dataset contains 392,692 valid transaction lines from 4,338 unique customers. The ten insights below summarize the most business-relevant patterns uncovered during EDA.

Insight 1 — Revenue Is Concentrated in a Small Set of Products

The top 10 products by revenue account for roughly 20% of total revenue, a clear Pareto (80/20) pattern. Best-sellers such as decorative and gift items dominate the top ranks, meaning stockouts on these SKUs carry outsized revenue risk.

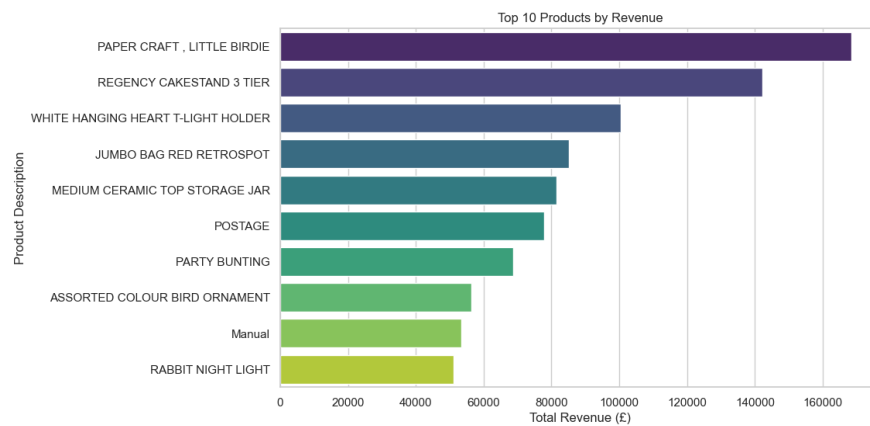


Figure 1. Top 10 products by total revenue.

Insight 2 — A Small Group of Customers Drives Disproportionate Revenue

The top 10 customers by total spend each contribute revenue far above the average customer, with the single largest account alone generating over £77,000 in the observed period — a strong early indicator that a wholesale or bulk-buying segment exists within an ostensibly retail dataset.

Insight 3 — Sales Are Strongly Seasonal, Peaking in November

Monthly revenue climbs steadily through the year and peaks sharply in November, consistent with pre-holiday and Christmas gift-buying behavior. This has direct implications for inventory build-up, staffing, and the timing of promotional campaigns.



Figure 2. Monthly revenue trend, December 2010 – December 2011.

Insight 4 — Weekday Trading Pattern, No Weekend Saturday Activity

Transaction volume is heaviest on Thursdays, with a gradual decline toward the end of the week; notably, the dataset shows no recorded Saturday transactions, indicating the retailer's order-processing (or the underlying B2B customer base) is not active that day. Sunday volume is also comparatively low.

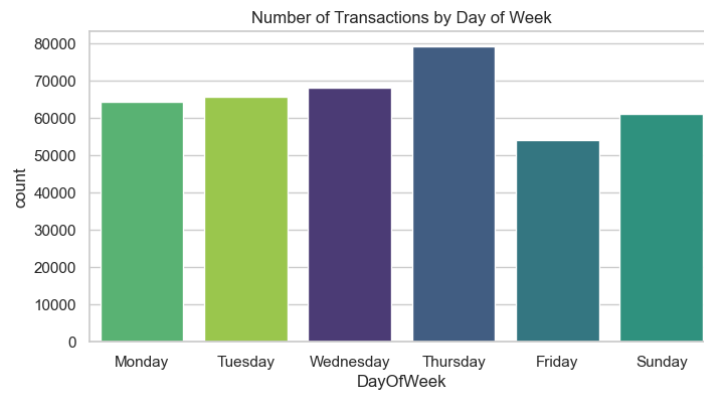


Figure 3. Number of transactions by day of week.

Insight 5 — Purchases Cluster Around Standard Business Hours

Transaction activity is concentrated between roughly 10:00 and 15:00, peaking around midday, and is minimal outside standard business hours. This is another signal consistent with a customer base that includes business buyers operating on a working schedule rather than pure end-consumer browsing at night.

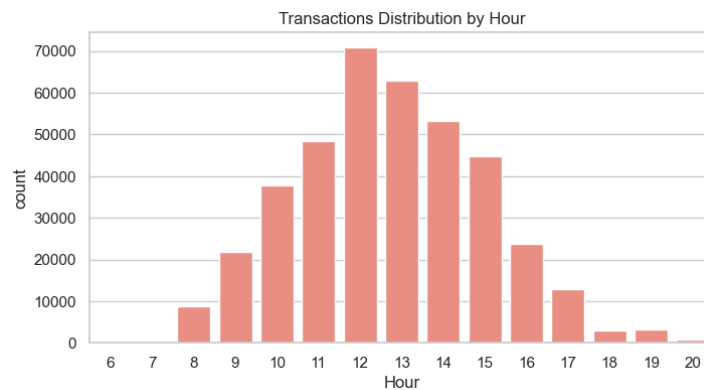


Figure 4. Transaction distribution by hour of day.

Insight 6 — The United Kingdom Dominates Revenue Geography

The United Kingdom is, by a wide margin, the largest revenue-generating market. The next four countries in the top-five ranking trail substantially but still represent a meaningful base of international demand that is likely under-served relative to its potential.

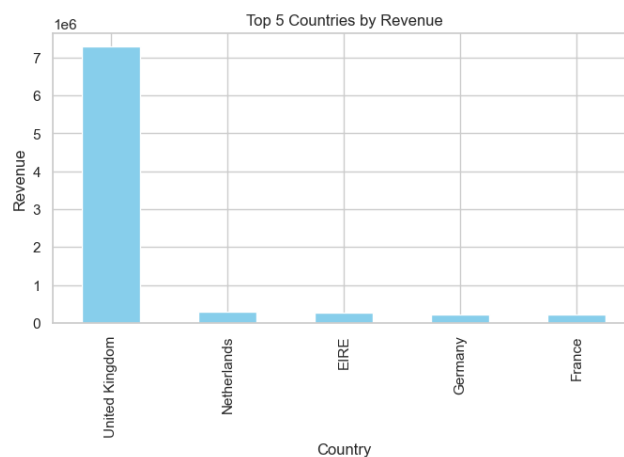


Figure 5. Top 5 countries by total revenue.

Business implication: the majority of revenue is generated from the UK market, and there is a potential opportunity to expand marketing or logistics investment in the top three high-performing international markets.

Insight 7 — Customer Value Distributions Are Heavily Skewed

Prior to any transformation, the Frequency and Monetary distributions are extremely right-skewed (skewness of 12.07 and 19.34 respectively), meaning the vast majority of customers make a small number of purchases while a small minority transact very frequently and spend very heavily. Recency is more moderately skewed (1.25).

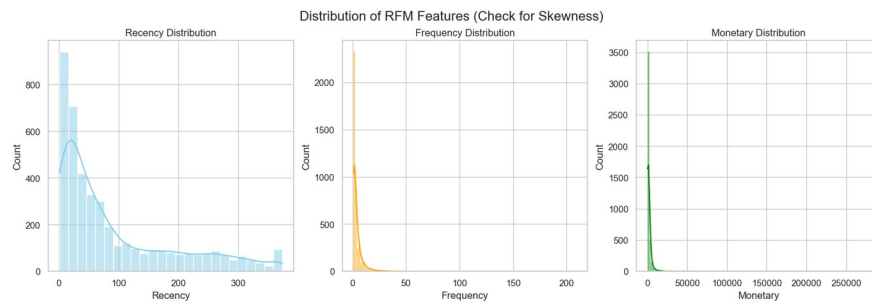


Figure 6. Distribution of Recency, Frequency, and Monetary values prior to transformation.

Insight 8 — Log Transformation Substantially Normalizes the Data

Applying a $\log(1+x)$ transformation to the RFM features sharply reduces skewness — Frequency falls from 12.07 to 1.21 and Monetary from 19.34 to 0.40 — confirming that the underlying behavior is exponential in nature (most customers buy rarely, a few buy constantly) and that transformation is an appropriate step to consider when preparing the data for distance-based algorithms.

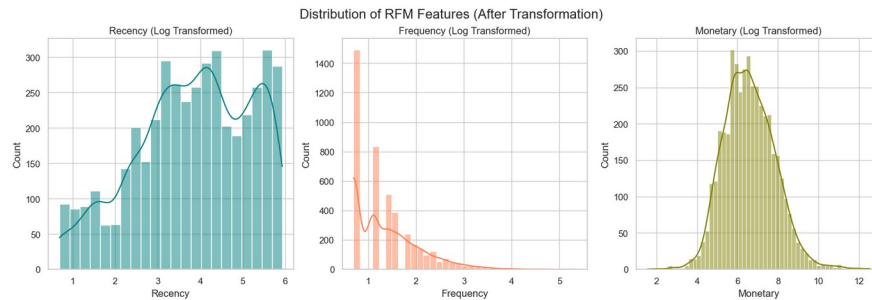


Figure 7. Distribution of Recency, Frequency, and Monetary values after log transformation.

Insight 9 — High-Value Outliers Represent a Distinct Population, Not Noise

Filtering Monetary value to the 5th–95th percentile range removes 434 customers (about 10% of the customer base) from the extremes. Rather than discard these as data-quality noise, the analysis treats them as a real and commercially important population — they reappear later as the core of the “Champions” segment.

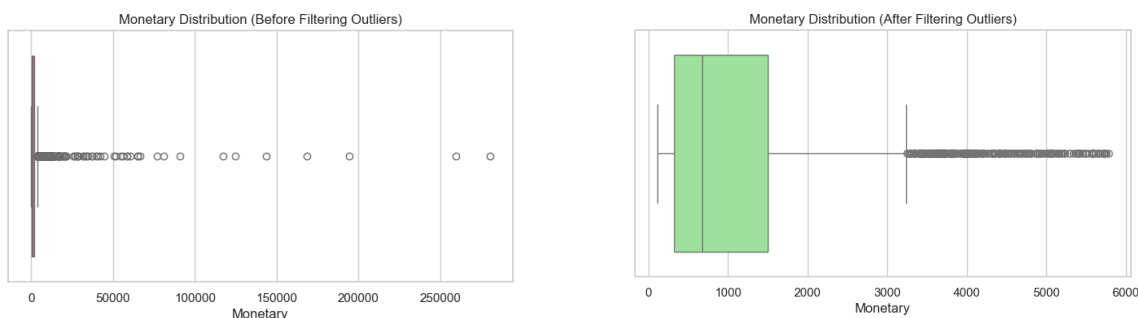


Figure 8. Monetary distribution before outlier filtering (left) and after 5th–95th percentile filtering (right).

Insight 10 — The Customer Base Naturally Separates into Three Behavioral Tiers

When Recency, Frequency, and Monetary are combined and clustered, three clearly differentiated groups emerge — ranging from customers who purchased very recently, very frequently, and at very high value, down to customers who have not purchased in a long time, rarely, and at low value. This structure, detailed in Section 3 and Section 4, forms the backbone of the segmentation strategy.

3. Customer Segmentation Methodology

3.1 RFM Feature Engineering

Customer-level behavior was summarized using the RFM framework, a widely used and interpretable approach in retail and CRM analytics:

- **Recency (R):** Number of days between the customer's most recent purchase and a fixed snapshot date (one day after the last transaction in the dataset). Lower values indicate more recently active customers.
- **Frequency (F):** Number of unique invoices (distinct purchase occasions) placed by the customer. Higher values indicate more habitual, repeat purchasing behavior.
- **Monetary (M):** Total revenue (Quantity × Unit Price, summed across all transactions) contributed by the customer. Higher values indicate greater cumulative commercial value.

These three metrics were computed at the CustomerID level, producing one row per customer (4,338 customers in total) with three numeric features.

3.2 Data Scaling

Because Recency, Frequency, and Monetary are measured on very different scales (days versus counts versus currency) and K-Means relies on Euclidean distance, the RFM features were standardized using scikit-learn's StandardScaler, which rescales each feature to zero mean and unit variance. This ensures that no single metric (for example, Monetary, which can range into tens of thousands of pounds) dominates the distance calculation purely due to its numeric scale.

The skewness analysis in Section 2 also confirmed that Frequency and Monetary are highly right-skewed; a log transformation was evaluated as a normalization step and shown to substantially reduce this skew, supporting the robustness of the distance-based clustering approach used on the standardized RFM features.

3.3 Choosing K-Means with 3 Clusters

K-Means was selected as the clustering algorithm for its interpretability, computational efficiency at this data volume, and its established track record in RFM-based customer segmentation. The optimal number of clusters was determined using two complementary diagnostics:

- **Elbow Method:** Inertia (within-cluster sum of squares) was computed for K ranging from 1 to 9. The inertia curve shows a pronounced bend around K = 3, beyond which additional clusters yield rapidly diminishing reductions in inertia — the classic signature used to select the elbow point.

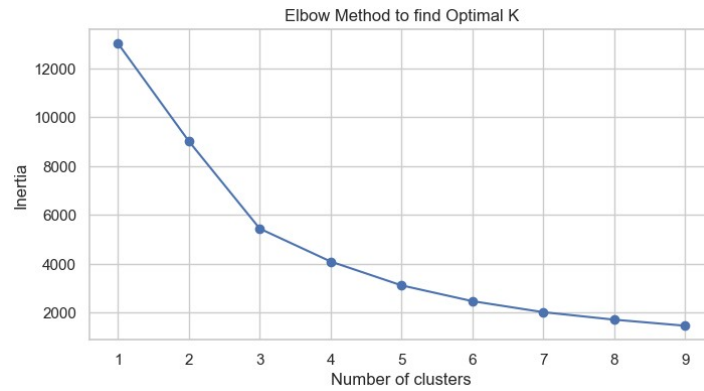


Figure 9. Elbow Method showing inertia versus number of clusters (K).

- **Silhouette Score:** With K = 3, the resulting clustering achieves a Silhouette Score of 0.594. This score, which ranges from -1 to 1, reflects how well each customer fits within its assigned cluster relative to neighboring clusters; a value of 0.594 indicates reasonably dense, well-separated, and meaningfully distinct clusters, supporting three segments as a statistically sound and business-interpretable choice.

Together, the Elbow Method and Silhouette Score confirm that three clusters strike the right balance between granularity (enough segments to drive differentiated action) and statistical robustness (each segment is cohesive and distinguishable from the others).

4. Cluster Interpretation

The table below summarizes the average RFM profile of each of the three clusters produced by the K-Means model. These averages are the primary evidence used to translate statistical clusters into named, business-meaningful customer segments.

Segment	Avg. Recency (days)	Avg. Frequency (orders)	Avg. Monetary (£)
Champions / VIP	6.0	66.4	85,826.08
Active / Loyal	41.5	4.7	1,849.67
At Risk / Hibernating	247.1	1.6	629.66

Table 1. Average Recency, Frequency, and Monetary value by customer segment.



Figure 10. Customer segments plotted by Recency versus Monetary value.

Champions (VIP Customers)

This is the smallest but by far the most valuable segment. On average, Champions purchased only about 6 days ago, have placed roughly 66 orders historically, and have generated an average of £85,826 in lifetime revenue — an order of magnitude above the other two segments. These are the retailer's most engaged, most frequent, and highest-spending customers, very likely including key wholesale or bulk-purchasing accounts. They represent the core of the business's revenue base and the highest priority for retention.

Active (Loyal, Mid-Value Customers)

Active customers purchased roughly 41 days ago on average, have made about 5 orders, and have spent close to £1,850 to date. This segment is healthy and engaged, but transacts less frequently and at meaningfully lower value than Champions. They represent a stable revenue base with clear headroom to be grown into the Champions tier through targeted engagement.

At Risk (Hibernating Customers)

At Risk customers have not purchased in an average of 247 days (over eight months), have historically placed just 1–2 orders, and have spent an average of only £630. This is the largest segment by customer count and represents both the greatest churn risk and the greatest reactivation opportunity: even a modest win-back rate applied to this large population could meaningfully offset revenue lost to churn.

5. Strategic Recommendations

The following segment-specific actions translate the clustering results into a practical marketing and retention roadmap.

Champions — Protect and Reward

Recommendation	Rationale / Expected Impact
VIP loyalty tier	Formal recognition (dedicated account manager, priority support, early access to new stock) reduces the risk of losing the retailer's highest-value accounts to competitors.
Exclusive perks & previews	Early access to new product lines and limited-run items rewards loyalty and reinforces habitual purchase frequency.
Proactive account management	Given their outsized average spend (£85,826), even small increases in retention translate into large absolute revenue protection; personal outreach is justified.
Referral incentives	Champions are natural advocates; structured referral rewards can convert their satisfaction into new customer acquisition at low cost.

Active — Nurture and Grow

Recommendation	Rationale / Expected Impact
Loyalty / points program	A structured rewards program incentivizes the additional purchase frequency needed to migrate Active customers toward the Champions profile.
Cross-sell and upsell campaigns	Personalized product recommendations based on past purchases can increase basket size and order frequency.
Milestone and anniversary offers	Time-based triggers (e.g., first-purchase anniversary) re-engage customers before natural recency drift begins.
Free-shipping or spend thresholds	Modest incentives to cross a spending threshold nudge Active customers toward higher Monetary value without heavy discounting.

At Risk — Win Back or Confirm Churn

Recommendation	Rationale / Expected Impact
Win-back email campaigns	Given an average recency of 247 days, targeted “we miss you” campaigns with a time-limited incentive are a proven lever to re-activate dormant accounts.

Recommendation	Rationale / Expected Impact
Discount or reactivation offers	A meaningful (but margin-aware) one-time discount can be the deciding factor for a lapsed customer to return, especially against their historically low average order value (£630).
Customer feedback surveys	Understanding why this large segment disengaged (price, product fit, service) provides root-cause input to prevent future churn from the Active segment.
Suppression / cost discipline	For customers who do not respond after repeated win-back attempts, reducing marketing spend directed at them protects campaign ROI and email sender reputation.

6. Conclusion

This analysis demonstrates that a relatively small set of levers — clean transaction data, a well-established RFM framework, and a statistically validated three-cluster K-Means model (Silhouette Score 0.594) — can convert over half a million raw transaction records into a clear, actionable customer segmentation. Champions, Active, and At Risk customers each warrant a distinct commercial motion: protect and reward the first, nurture and grow the second, and win back or responsibly suppress the third. Combined with the seasonal, product, and geographic insights from the exploratory analysis, this segmentation provides a practical foundation for the retailer's next marketing planning cycle.